

12. Page Replacement using FIFO

```
#include <stdio.h>

int main () {
    int n,j,i,frames,k=0,faults=0;

    printf("Enter the number of pages: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter page reference string: ");
    for (i=0;i<n;i++) {
        scanf("%d",&pages[i]);
    }

    printf("Enter the number of frames: ");
    scanf("%d", &frames);

    int frame[frames];

    for (i=0;i<frames;i++) {
        frame[i] = -1;
    }

    for (i=0;i<n;i++) {
        int found = 0;

        for (j=0;j<frames;j++) {
            if (frame[j] == pages[i]) {
                found = 1;
                break;
            }
        }

        if (!found) {
            frame[k] = pages[i];
            k = (k+1) % frames;
            faults++;
        }
    }

    printf("Total page faults: %d\n", faults);

    return 0;
}
```